

GC University Lahore

Department of Computer Science

Examination Fall 2021

Course Title : Computer Organization and Assembly Language	Course Code: CS-2108 Time: 150 Minutes
Total Marks: 50	Name: _____ Roll #: _____

Instructions:

1. Read questions carefully before attempting.
2. Question No. 1 is compulsory.
3. Calculators are not allowed.

Q. No. 1: (10)

Relate C Code and Equivalent Assembly Code also Draw stack frame. Your stack should be clearly representing all steps.

C Code	Assembly Code
<pre> void rreverseArray(int arr[], int start, int end) { while (start < end) { int temp = arr[start]; arr[start] = arr[end]; arr[end] = temp; start++; end--; } } int sender() { int A[10] = {5,6,7,9,1,0,10,11,2,3}; rreverseArray(A,0, 9); return 0; } </pre>	<pre> _rreverseArray: pushl %esi pushl %ebx movl 12(%esp), %ecx movl 16(%esp), %edx movl 20(%esp), %eax cmpl %eax, %edx jge L1 L3: movl (%ecx,%edx,4), %ebx movl (%ecx,%eax,4), %esi movl %esi, (%ecx,%edx,4) movl %ebx, (%ecx,%eax,4) addl \$1, %edx subl \$1, %eax cmpl %eax, %edx jl L3 L1: popl %ebx popl %esi ret _sender: subl \$60, %esp movl \$5, 20(%esp) </pre>

	movl \$6, 24(%esp)
	movl \$7, 28(%esp)
	movl \$9, 32(%esp)
	movl \$1, 36(%esp)
	movl \$0, 40(%esp)
	movl \$10, 44(%esp)
	movl \$11, 48(%esp)
	movl \$2, 52(%esp)
	movl \$3, 56(%esp)
	movl \$9, 8(%esp)
	movl \$0, 4(%esp)
	leal 20(%esp), %eax
	movl %eax, (%esp)
	call _rverseArray
	movl \$0, %eax
	addl \$60, %esp
	ret

Q. No. 2: (2x5=10)

A. Answer Following Questions in precise way (given time constraint avoid unnecessary details).

- What are special numbers in IEEE-754 standard, Explain all cases in special numbers?
- Why switch statement is more efficient than if-else structure?
- What are CISC and RISC architecture in terms of Instruction Set Architecture?
- Explain the functionality of LEAL instruction?
- What are the callee and caller registers in IA-32, what happens when callee uses the caller register?

Q. No. 3: (5+5)

A. Explain, Fetch, Decode and Execute Stages of Y-86 processor with the help of diagram.

B. In the IA-32 processor, Explain following

- The role of PC (%EIP) in code.
- The functionality of %ESP in stack area.
- Which registers are used in full multiplication instruction, write exact format?
- Which registers are used in full division instruction, write exact format?
- Role of %CL in shift instructions

Q. No. 4: (6+4)

A. Assume 6-bit floating point number, 1-sign bit, k=3 bit for exponent and n=2 bit for mantissa. With the bias ($2^{k-1} - 1$).

your task is to fill in at least 05 more entries anywhere in the table with must entries of **Zero**, **smallest positive number**, **Largest denormalized number**, **Smallest Normal number**, **One**, **Largest normal number** and **Infinity**. (Write your entries in fractional format)

Description	Bit Representation	Exponent			Fraction		Value	
		c	E	2^E	f	M	$2^E \times M$	V

B. Consider Following Object Code and exe Code, your task is to fill in the offset in object code and virtual addresses in exe code, also compute the jump encoding at (????????) for forward and backward jumps in object and exe code. The starting offset of object is 0 and start virtual address of exe is 401550.

Object Code

Offset	Machine Code	Assembly code
0	8b 54 24 04	mov 0x4(%esp),%edx
4	83 fa 01	cmp \$0x1,%edx
7	7e 11	jle ????????
9	b8 01 00 00 00	mov \$0x1,%eax
E (14)	0f af c2	imul %edx,%eax
11 (17)	83 ea 01	sub \$0x1,%edx
14 (20)	83 fa 01	cmp \$0x1,%edx
17 (23)	75 f5	jne ????????
19 (25)	c3	ret
1A (26)	b8 01 00 00 00	mov \$0x1,%eax
1F (31)	c3	ret

Exe Code

Virtual Add	Machine Code	Assembly Code
	83 f9 01	cmp \$0x1,%ecx
	7e 11	jle ??????????????
	b8 01 00 00 00	mov \$0x1,%eax
	0f af c1	imul %ecx,%eax
	83 e9 01	sub \$0x1,%ecx
	83 f9 01	cmp \$0x1,%ecx
	75 f5	jne ??????????????
	c3	retq
	nop	

Q No. 5: (5+5)

A. Write a brief note on Condition Codes/ Flags. Which instructions affect flags and which do not affect flags. What is the role of CMP and TEST instructions in relation with Condition Codes in assembly programming?

B. Write a detailed note on Conditional Move Instruction. In which scenario they are applicable and how they are efficient from control transfer instruction. Elaborate your answer with a comparative example in C/ Assembly Code.

Q. No. 6: (5+5)

A. Consider following assembly code, Write equivalent C code. The header for c code is
int function(int x, int y)

_function:

```
movl 4(%esp), %eax → x
leal 5(%eax, %eax, 4), %eax → 5x
addl 8(%esp), %eax → result = x + y;
sall $4, %eax → result = result * 4;
xorl $0xffffffff, %eax →
movbl %al, %eax → return result;
```

B. What are operand specifier methods/ addressing modes in IA-32. Explain uses of addressing modes.
